



EAST WEST UNIVERSITY

Lab Report-5

Submitted By:

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Section: 01

Submitted To:

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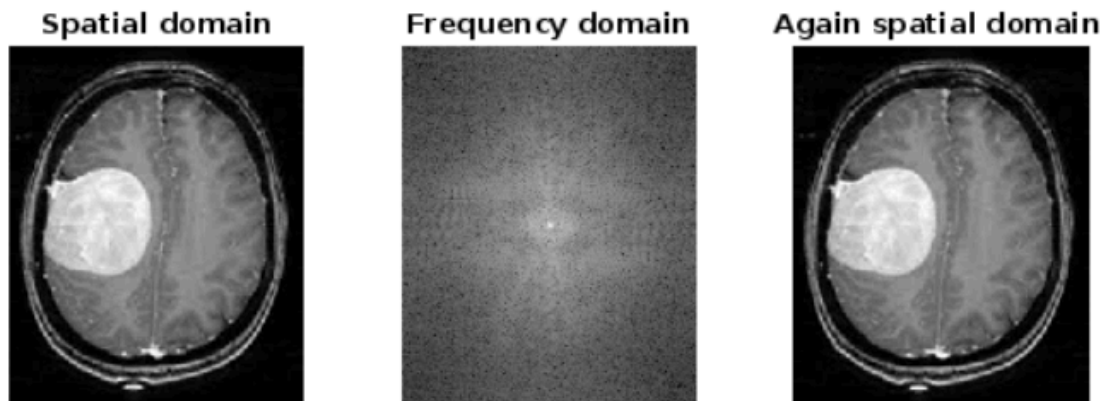
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1. Apply Fourier transform to transform any image (above) from the spatial domain to the frequency domain. Apply inverse Fourier transform to transform the image from the frequency domain to the spatial domain.

Answer:

```
img = imread('picture_5.1.jpg');
%convert rgb image to gray image
gray_img = rgb2gray(img);
%fourier transform of the image
freq_domain=fft2(double(gray_img));
freq_domain_shifted= fftshift(freq_domain);
magnitude_spectrum= log(1+ abs(freq_domain_shifted));
%inverse
freq_domain_shifted_inverse=ifftshift(freq_domain_shifted);
img_back= real(ifft2(freq_domain_shifted_inverse));
%figure
subplot(1,3,1), imshow(img); title('Spatial domain');
subplot(1,3,2), imshow(magnitude_spectrum, []); title('Frequency domain');
subplot(1,3,3), imshow(img_back, []); title('Again spatial domain ');
```

Output:



2. Apply three types of high pass filtering in the frequency domain in **Figure 1** and find out which one is better to produce the enhanced image (sharpen) for the given image (output must show all steps as shown in **Figure 2**).

- i. Ideal high pass filter (IHPF)
- ii. Butterworth high pass filter (BHPF)
- iii. Gaussian high pass filter (GHPF)

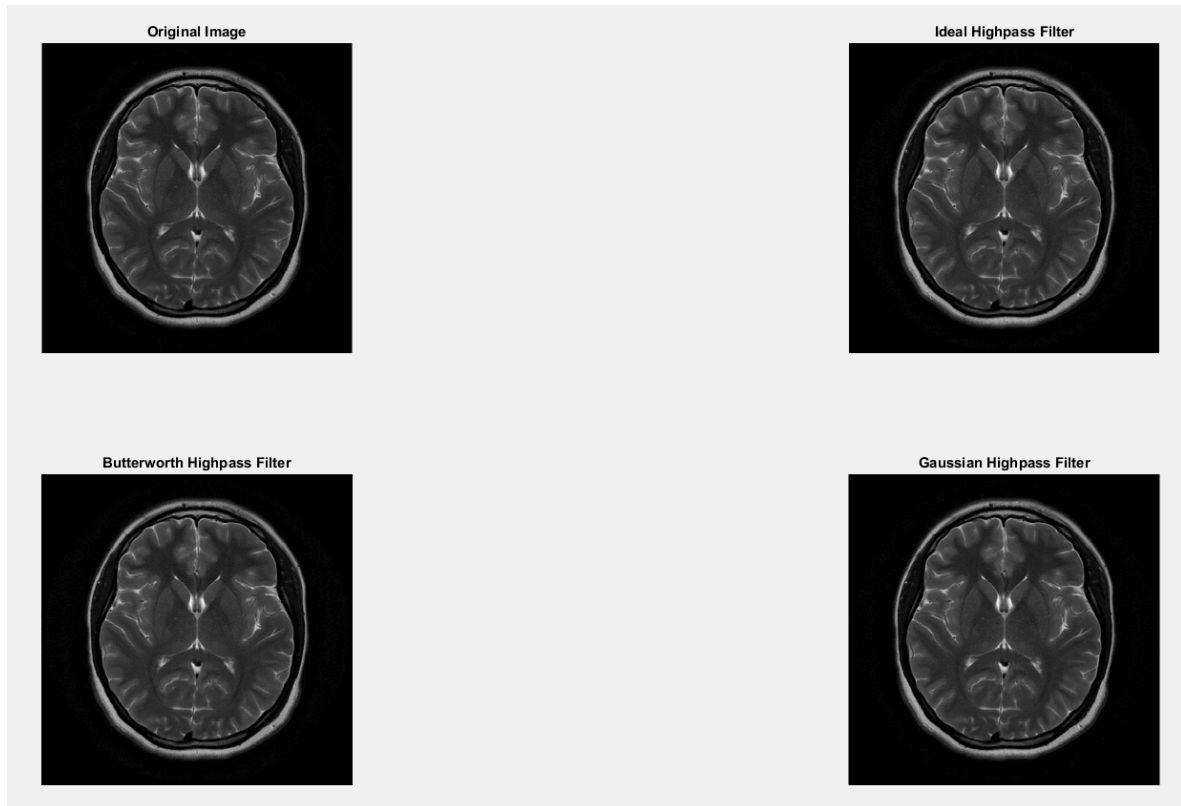
Solution:

```

img = imread('picture_5.4.jpg');
gray_img=img
% height and weight of the image
[a,b] = size(gray_img);
% Getting Fourier Transform of the image using fft2
ft_img = fft2(double(img));
% Designing filter
u = 0:(a-1);
idx = find(u>a/2);
u(idx) = u(idx)-a;
v = 0:(b-1);
idy = find(v>b/2);
v(idy) = v(idy)-b;
% MATLAB library function meshgrid(v, u) returns 2D grid
% Matrix V with each row is a copy of v, and matrix U
% with each column is a copy of u
[V, U] = meshgrid(v, u);
D0 = 70; %Cutoff frequency of IHPF
n = 20; %order of butterworth filter
D1= 70; %Cutoff frequency of Butterworth filter
sigma=20; %Cutoff frequency of Butterworth filter
% Calculating Euclidean Distance
D = sqrt(U.^2+V.^2);
% Comparing with the cut-off frequency and determining the filtering mask
H1= double(D > D0);
% Convolution between the Fourier Transformed image and the mask for IHPF
G1 = ft_img.*H1;
% determining the filtering mask
H2 = 1./(1 + (D1./sqrt(D)).^(2*n));
% Convolution between the Fourier Transformed image and the mask for BHPF
G2= ft_img.*H2;
% Comparing with the cut-off frequency and determining the filtering mask
H3= 1-exp(-(U.^2+ V.^2)./(2*sigma^2));
% Convolution between the Fourier Transformed image and the mask for GHPF
G3= ft_img.*H3;
%shift the zero frequency component back to the corners
G1=ifftshift(G1);
G2=ifftshift(G2);
G3=ifftshift(G3);
% Getting the resultant image by IFT, butter
% of the convoluted image using MATLAB library function
% ifft2 (2D inverse fast fourier transform)
output_image_IHPF = real(ifft2(double(G1)));
output_image_BHPF = real(ifft2(double(G2)));
output_image_GHPF = real(ifft2(double(G3)));
% Displaying Input Image and Output Image
subplot(2, 2, 1), imshow(img), title('Original Image');
subplot(2, 2, 2), imshow(output_image_IHPF, [ ]), title('Ideal Highpass
Filter');
subplot(2, 2, 3), imshow(output_image_BHPF, [ ]), title('Butterworth
Highpass Filter');
subplot(2, 2, 4), imshow(output_image_GHPF, [ ]), title('Gaussain
Highpass Filter');

```

Output:



Explanation: Among these three images butterworth high pass filter is give more sharper image than others. And ideal high pass filter and gaussian high pass filter is less sharpen then Butterworth high pass filter.

3. Apply three types of low pass filtering in the frequency domain in **Figure 1** and find out which one is better to produce the smoothen image for the given image (output must show all steps as shown in **Figure 2**).

- i. Ideal low pass filter (ILPF)
- ii. Butterworth low pass filter (BLPF)
- iii. Gaussian low pass filter (GLPF)

Solution:

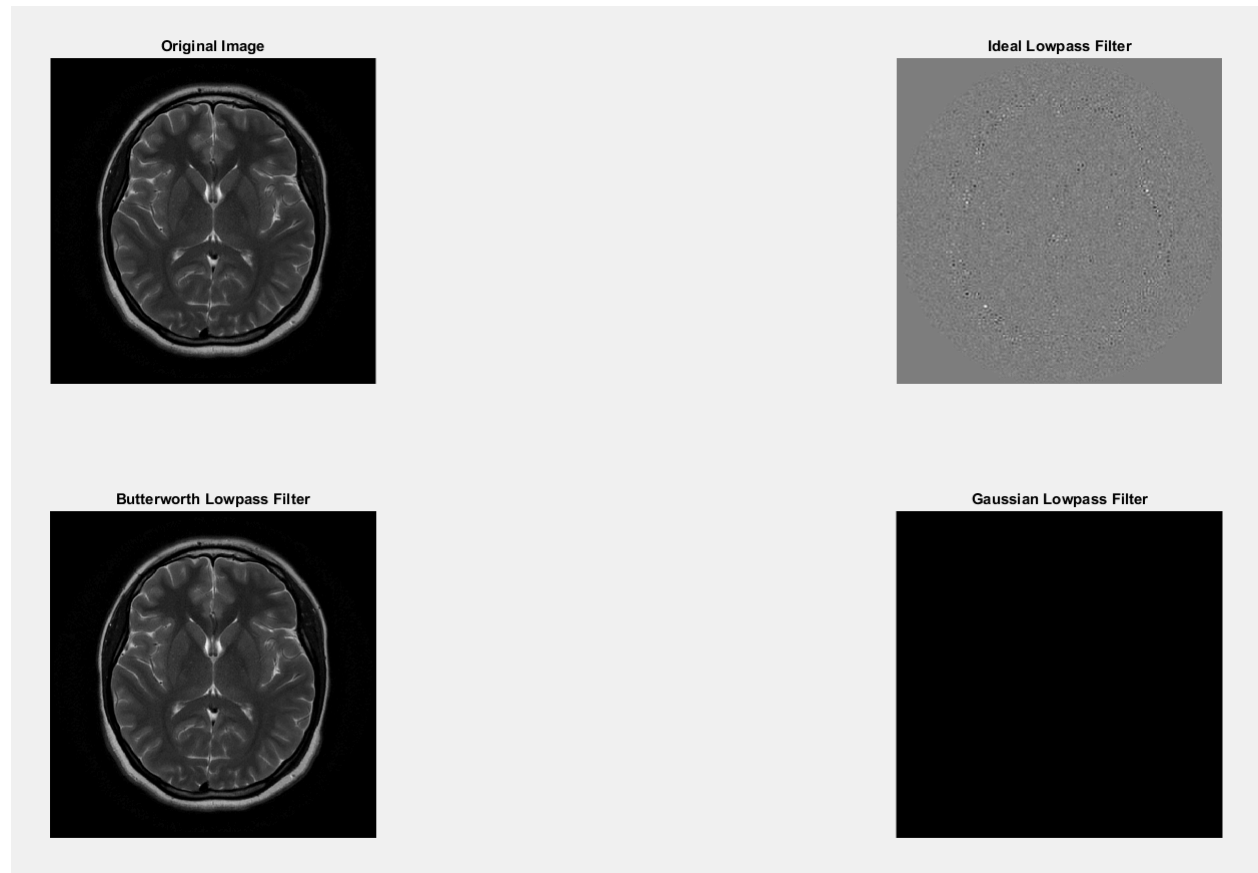
```
img = imread('picture_5.4.jpg');  
img_gray = img;  
% Take the Fourier transform of the image  
img_fft = fft2(double(img_gray));  
% Compute the row and column size of the image  
[a,b] = size(img_gray);
```

```

% cutoff frequency for the filters
D0 = 100;
% Define the order for the Butterworth filter
n = 2;
% Define the standard deviation for the Gaussian filter
sigma = 10;
% Apply ideal lowpass filter
H_ilpf = zeros(a,b);
for i = 1:M
    for j = 1:N
        if sqrt((i-a/2)^2 + (j-b/2)^2) <= D0
            H_ilpf(i,j) = 1;
        end
    end
end
img_ilpf = real(ifft2(img_fft .* H_ilpf));
% Apply Butterworth lowpass filter
H_blpf = 1 ./ (1 + (sqrt(2)-1)*(sqrt((i-M/2)^2 + (j-N/2)^2)/cutoff_freq)^(2*order));
img_blpf = real(ifft2(img_fft .* H_blpf));
% Apply Gaussian lowpass filter
H_glpf = exp(-((i-M/2)^2 + (j-N/2)^2)/(2*sigma^2));
img_glpf = real(ifft2(img_fft .* H_glpf));
% Display images
subplot(2,2,1); imshow(img_gray); title('Original Image');
subplot(2,2,2); imshow(img_ilpf, []); title('Ideal Lowpass Filter');
subplot(2,2,3); imshow(img_blpf, []); title('Butterworth Lowpass Filter');
subplot(2,2,4); imshow(img_glpf, []); title('Gaussian Lowpass Filter');

```

Output:



Explanation: Among these three butterworth low pass filter is give more smoothness then others. Ideal low pass is not clearly visible and gaussian is totally invisible.